

Yangon Tech Hub Development & Coding Standards

This document defines the official coding standards, development workflow, naming conventions, and engineering practices for the Yangon Tech Hub ERP platform development team. These rules focus only on software development and coding processes.

1. Frontend Development Standards

- Frontend framework must use Next.js with TypeScript.
- Use reusable components whenever possible.
- Responsive design is required for desktop and mobile devices.
- Avoid inline CSS styles.
- Use modular or utility-based styling approach.

2. Backend Development Standards

- Backend services must use Spring Boot REST APIs.
- Use layered architecture (Controller, Service, Repository).
- All APIs must include validation and error handling.
- JWT authentication and RBAC are mandatory.
- Swagger/OpenAPI documentation must be maintained.

3. Database Standards

- MySQL is the official project database.
- All tables must include created_at and updated_at columns.
- Use foreign keys and indexes properly.
- Use snake_case for database tables and columns.
- Sensitive data must never be stored as plain text.

4. Variable & Function Naming

- Use camelCase for variables and function names.
- Variable names must be meaningful and readable.
- Avoid short unclear names like a, b, temp.
- Examples: userProfile, calculateSalary(), attendanceStatus.

5. Constant Naming Convention

- Use UPPER_SNAKE_CASE for constants.
- Examples: API_BASE_URL, MAX_LOGIN_RETRY.

6. File & Component Naming

- Use PascalCase for React/Next.js component files.
- Use kebab-case for utility and service files.
- Examples: EmployeeCard.tsx, attendance-service.ts.

7. Folder Structure Rules

- Frontend must be organized by modules/features.
- Recommended folders: components/, services/, hooks/, pages/, utils/.
- Avoid deeply nested folder structures.

8. API Naming Convention

- REST API endpoints must use lowercase and plural nouns.
- Examples: /api/employees, /api/projects.
- Avoid verbs inside endpoint names.

9. DTO & Entity Naming

- Use clear suffixes for DTOs and entities.
- Examples: EmployeeResponse, CreateUserRequest, UserEntity.

10. Git Branch Naming

- Use feature/* for features.
- Use bugfix/* for bug fixes.
- Use hotfix/* for urgent fixes.
- Use release/* for release preparation.

11. Git Commit Standards

- Commit messages must be clear and meaningful.
- Use prefixes: feat:, fix:, refactor:, docs:, chore:.
- Example: feat: add attendance API endpoint.

12. Security Development Rules

- Passwords must always be encrypted.
- All APIs must validate authentication tokens.
- Prevent SQL Injection and XSS vulnerabilities.
- Environment variables must never be hardcoded.

13. Code Quality Standards

- Use ESLint and Prettier for code formatting.
- Functions should be small and single-purpose.
- Avoid duplicated logic.
- Write readable and maintainable code.

14. Development Workflow Rules

- All tasks must be tracked in the project board.
- Feature branches are required before implementation.
- Pull Requests are required before merging.
- Code review approval is mandatory.

Project: Smart ERP & Workforce Management Platform

Client: Yangon Tech Hub

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